

Hiring Assessment

MISN Lab, IIT Delhi

AdiVaani Initiative

Due Date: May 25, 2026, 12:00 PM IST

1 Overview

The **AdiVaani Initiative** at **MISN Lab, IIT Delhi** focuses on developing multilingual and low-resource NLP systems for Indian languages, with particular emphasis on indigenous and underrepresented language communities. This assessment is intended to evaluate your understanding of deep learning fundamentals, modern NLP systems, architectural reasoning, experimentation methodology, implementation quality, and engineering maturity.

The objective of this assignment is not merely obtaining the highest benchmark score. We are significantly more interested in how you reason about problems, how you justify architectural choices, how you design experiments, how rigorously you analyze both successful and unsuccessful outcomes, and whether you follow sound engineering and system design principles while building reproducible, maintainable, and well-structured machine learning systems.

A Hindi–Marathi parallel corpus will be provided for this assignment.

Please download the dataset from here

<https://shorturl.at/9Zkkn>

2 General Instructions

- You must use your own compute resources to complete this assignment. Your implementation should be capable of efficiently utilizing available GPU resources wherever applicable.
- You are free to preprocess, tokenize, clean, augment, filter, or restructure the dataset in any manner you believe is appropriate for the task. Using the full dataset is also not necessary. You can use partial dataset by randomly sampling points.
- You may pretrain models multiple times if required.
- You are allowed to use LLMs for code assistance or experimentation. However, every such usage must be explicitly documented in your report. You must thoroughly explain and justify every architectural, optimization, and implementation decision. During the interview round, you will be expected to independently defend all aspects of your submitted code and reasoning.
- A technical interview round will follow the submission process. The discussion may involve your implementation details, deep learning fundamentals, optimization methods, Transformer architectures, training dynamics, NLP concepts, mathematical reasoning, and design trade-offs involved in your system.

- You are strictly prohibited from using readily available pretrained Seq2Seq translation systems such as mT5, NLLB-200 variants, MarianMT, M2M-100, or any directly available machine translation framework.
- Although translation quality and evaluation metrics will naturally be considered, accuracy alone is not the sole evaluation criterion. Considerable importance will be given to your experimental methodology, engineering quality, architectural understanding, ability to reason about failures, and overall technical clarity.
- Your submission must include a detailed technical report discussing dataset processing methodology, tokenization strategy, architecture design, optimization decisions, hyperparameter choices, ablation studies, training observations, failure analysis, computational constraints, and experimental findings. You must also clearly mention the GPU hardware used for training and any LLM systems used during development.
- Your experiments must include:
 - Train and validation loss plots,
 - Train and validation BLEU-100 plots,
 - Train and validation CHRF++-100 plots.
- The final submission must consist of a **public GitHub repository** containing your complete implementation, reproducibility instructions, training/evaluation scripts, and all relevant configurations required to reproduce your experiments. We will share a google form in due time.
- **Submission Deadline: May 25, 2026, 12:00 PM IST**
- No extensions will be granted under any circumstances.

3 Part I – Classical Neural Machine Translation

3.1 Objective

In this section, you are required to design and implement a complete Neural Machine Translation (NMT) framework for Hindi $\leftrightarrow$ Marathi translation using recurrent neural architectures.

3.2 Task Description

You must build a Seq2Seq translation framework consisting of an LSTM encoder, an LSTM decoder, with a **Attention** mechanism. Beyond these requirements, you are free to make architectural and engineering decisions regarding the decoding strategy, beam search methodology, optimization pipeline, tokenization strategy, curriculum learning setup, or training dynamics.

A major objective of this section is to evaluate how classical sequence modeling architectures behave under different representation learning settings.

You are required to perform experiments under at least the following embedding configurations:

1. Randomly initialized embeddings on both source and target sides.
2. BERT-based embeddings on both source and target sides using the following pretrained models:

Hindi BERT:

<https://huggingface.co/l3cube-pune/hindi-bert-v2>

Marathi BERT:

<https://huggingface.co/l3cube-pune/marathi-bert-v2>

You are encouraged to explore additional ideas such as shared embeddings, SentencePiece/BPE tokenization, subword vocabulary sharing, frozen versus trainable embeddings, cross-lingual embedding alignment, scheduled sampling, label smoothing, or beam search decoding strategies if you believe they improve the system.

An important component of this section is the comparative analysis between randomly initialized embeddings and pretrained BERT-based embeddings. Your experiments and report must critically analyze how the two embedding strategies affect convergence behavior, optimization stability, semantic preservation, translation fluency, low-frequency word handling, and final translation quality.

3.3 Evaluation Requirements

Your experiments must include BLEU (100) and CHRF++ (100) evaluations, where the scores are reported on a scale from 0 to 100. In addition, you must provide train and validation loss curves, train and validation BLEU (100) plots, and train and validation CHRF++ (100) plots.

Your report must contain a comparative quantitative discussion analyzing the differences between random embeddings and BERT-based embeddings across BLEU (100) and CHRF++ (100) scores. The discussion should clearly explain observed performance differences, strengths, limitations, convergence characteristics, and linguistic behavior under both embedding settings.

Your report should also contain qualitative analysis involving example translations and comparative observations between the different embedding configurations explored during experimentation.

3.4 Expected Deliverables

The final submission for this section should include complete training code, evaluation scripts, reproducible experiment configurations, plots, checkpoints, and a detailed technical discussion of the overall approach.

4 Part II – Language Model Pretraining and Translation

4.1 Objective

The second section of the assessment focuses on pretraining language models from scratch and exploring how these pretrained representations can be utilized for machine translation.

This section is intentionally designed to be research-oriented and open-ended. We are particularly interested in your reasoning process, creativity, architectural understanding, and your ability to connect ideas from modern language modeling literature to practical machine translation systems.

4.2 Pretraining Requirements

You are required to pretrain the following models entirely from scratch:

1. A BERT-like encoder model containing approximately 110M parameters. The NSP (Next Sentence Prediction) objective must be removed, and you are expected to design your own masked language modeling pipeline.
2. A GPT-2 style decoder-only model containing approximately 124M parameters.

Your Transformer implementations must incorporate the following architectural modifications:

- Rotary Positional Embeddings (RoPE) instead of sinusoidal positional embeddings,
- Grouped Query Attention (GQA) instead of standard multi-head attention,
- RMSNorm instead of LayerNorm.

4.3 Core Challenge

After pretraining these models, you must use them for machine translation.

How you choose to accomplish this is intentionally left open. You may pretrain more than one time or use any other principled methodology that you believe is technically sound.

The objective here is not simply reproducing an existing paper, but demonstrating your ability to reason about how pretrained language representations can be leveraged for translation tasks under practical constraints.

4.4 Hint

A BERT-style model fundamentally learns bidirectional contextual representations, whereas GPT-style models specialize in autoregressive generation.

Think carefully about how these complementary properties might be jointly exploited inside a translation framework.

You may also study existing architectural foundations such as:

`google-bert/bert-base-cased`

`openai-community/gpt2`

to better understand architectural philosophy, training dynamics, autoregressive decoding behavior, contextual representation learning, and modern language modeling principles.

You are encouraged to reason carefully about how encoder-style and decoder-style pre-trained representations differ, and how these differences may influence downstream translation quality, generation stability, and linguistic generalization.

4.5 Expected Discussion Areas

Your report should ideally discuss why your chosen architecture should work, what trade-offs are involved, the scaling limitations encountered during experimentation, optimization challenges, training stability issues, parameter efficiency considerations, failure cases, and possible future improvements.

We are particularly interested in whether you can critically analyze your own approach instead of merely presenting successful results.

5 Evaluation Philosophy

This assessment is deliberately research-oriented.

We are not only evaluating whether your models perform well, but whether you understand what you are building, whether you can reason rigorously about architectural decisions, whether you can engineer reproducible systems, and whether you can communicate technical ideas with clarity and precision.

6 Suggested References

1. Bahdanau et al., *Neural Machine Translation by Jointly Learning to Align and Translate*, ICLR, 2015.
2. Vaswani et al., *Attention Is All You Need*, NeurIPS, 2017.
3. Devlin et al., *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*, NAACL, 2019.
4. Radford et al., *Language Models are Unsupervised Multitask Learners*, OpenAI, 2019.
5. Su et al., *RoFormer: Enhanced Transformer with Rotary Position Embedding*, 2021.
6. Ainslie et al., *GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*, EMNLP, 2023.

7. Zhang and Sennrich, *Root Mean Square Layer Normalization*, NeurIPS, 2019.

Best of luck.